TRIVIAL KAN FIBRATIONS ARE STABLE UNDER BASE CHANGE

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-trivial-kan-base-change (source lines 5802-5807); the proof environment begins on the second line after it and is its own proof, closing at source line 5812.

Editorial note. The author's proof is one sentence referring to the functorial properties of the fibre product; the card reproduces it as written, carries the cited lemma as context, and records in the pairing note that no further detail is supplied by the author (nothing is silently completed).

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Methods chapter, counted once: a trivial Kan fibration remains a trivial Kan fibration after base change.

Allowed standard prerequisites (editorial).

- Trivial Kan fibrations. The horn-lifting property defining a trivial Kan fibration of simplicial sets.. It is the hypothesis and the conclusion (source lines 5804-5806).
- Fibre products of simplicial sets and their functoriality. The fibre product of simplicial sets, computed degreewise, with its functoriality in both variables (the author's lemma-fibre-product, retained as context).. It is what the author invokes for the base change (source lines 5810-5811).

Lemma 0.1. *If U, V, W are simplicial objects in the category $\mathcal{C}$, and if $a : V \to U$, $b : W \to U$ are morphisms and if $V \times_U W$ exists, then we have*

$$\mathrm{Mor}(T, V \times_U W) = \mathrm{Mor}(T, V) \times_{\mathrm{Mor}(T,U)} \mathrm{Mor}(T, W)$$

for any fourth simplicial object T of $\mathcal{C}$.

Proof. Omitted. □

Lemma 0.2. *Let $f : X \to Y$ be a trivial Kan fibration of simplicial sets. Let $Y' \to Y$ be a morphism of simplicial sets. Then $X \times_Y Y' \to Y'$ is a trivial Kan fibration.*

Date: stacks-project:simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Proof. This follows immediately from the functorial properties of the fibre product (Lemma 0.1) and the definitions. □